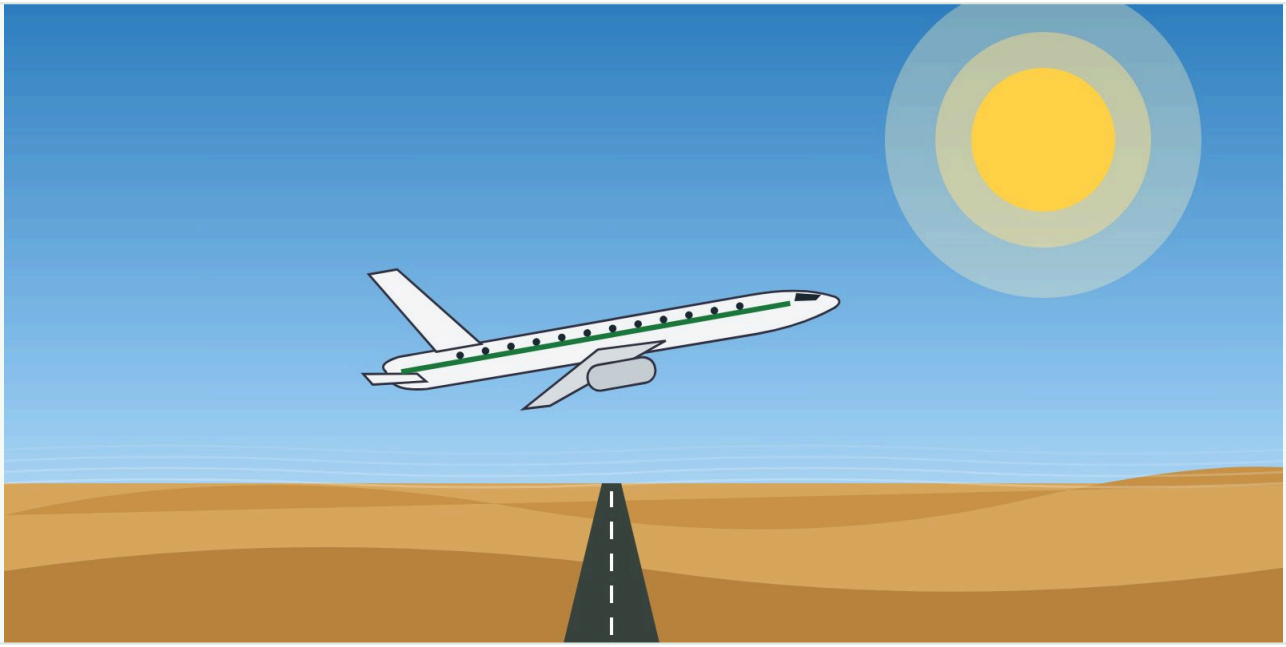
MODULE 1 · PERFORMANCE AND AIRCRAFT SYSTEMS

Lesson 01: Hot and High: Effects on Performance

Lesson 1 of 6 · about 8 min

LEARNING OBJECTIVES

- Explain how temperature and elevation affect aircraft performance.
- Identify the performance limits that become critical in hot weather.
- Use the performance tools correctly.

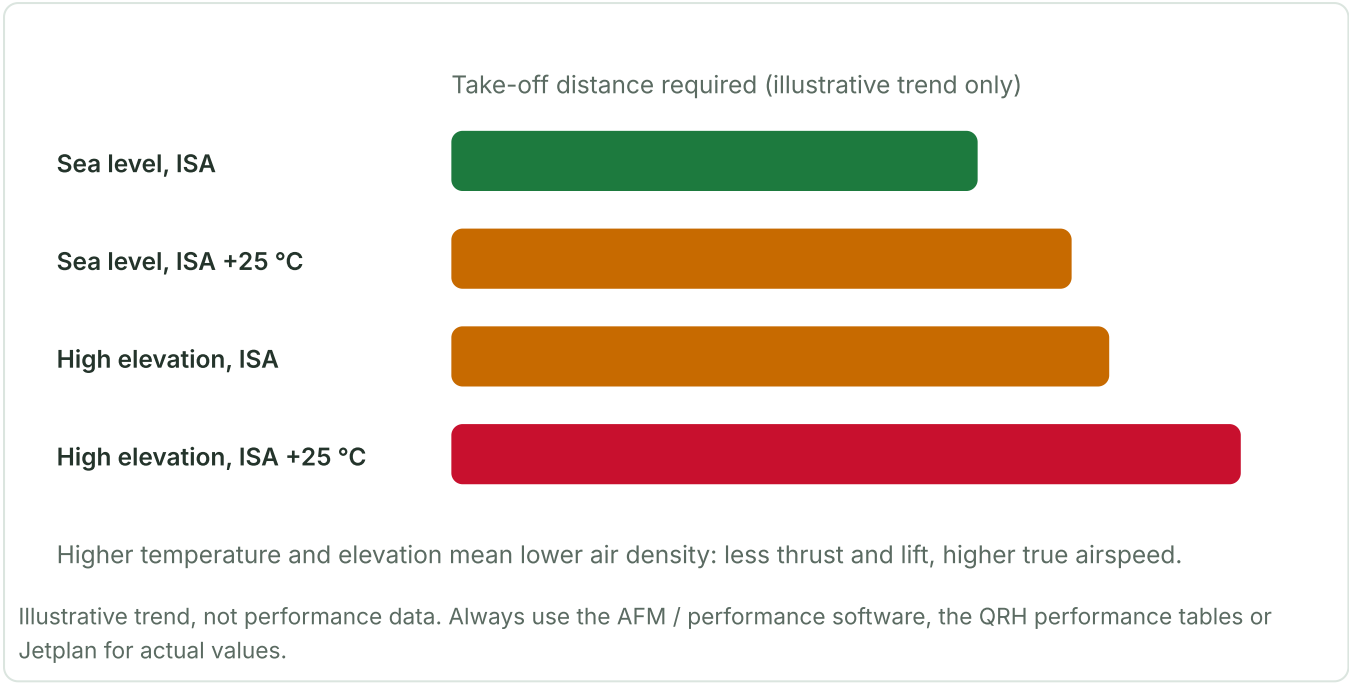


Hot and high: less density, less thrust, longer distances.

Density is everything

High temperature and high aerodrome elevation both reduce **air density** — expressed as a high **density altitude**. Lower density means:

- **less thrust or power** from the engines — jets and turboprops alike are flat-rated only up to a certain temperature;
- **less lift** at a given true airspeed — so the same indicated airspeed means a **higher true airspeed and ground speed**;
- longer take-off and landing distances, higher tyre speeds, and more energy for the brakes to absorb in a rejected take-off;
- reduced **climb gradient** — the climb limit (weight–altitude–temperature) and obstacle clearance often limit the take-off mass on hot days.



Illustrative trend, not performance data. Always use the AFM / performance software, the QRH performance tables or Jetplan for actual values.

Limits that become critical

- **Climb limit** and **obstacle limit** — especially at southern aerodromes with high elevation and temperatures.
- **Runway limit** — check NOTAMs for reduced declared distances (for example, a displaced threshold or the first part of a runway unusable).
- **Tyre speed limit** and **brake energy limit** — higher ground speeds at high density altitude.
- **Landing distance** — higher true airspeed at touchdown; a late or long flare uses runway very quickly.
- **Maximum temperature limits** for take-off and landing in the AFM.

Performance tools

- Use the **approved** performance data: AFM / company performance software, QRH performance tables and the Jetplan operational flight plan.
- Enter the **actual** temperature, QNH, wind and runway condition. A few degrees can change the limiting mass.
- Check aerodrome accessibility: take-off alternate, destination alternate and en-route alternates.
- The first officer prepares the performance dispatch; the captain cross-checks it.

KEY TAKEAWAYS

- Heat and elevation reduce density: less thrust and lift, higher true airspeed and ground speed.
- Climb, obstacle, tyre-speed and brake-energy limits become critical.
- Use approved performance data with actual conditions; check NOTAMs for reduced distances.

KNOWLEDGE CHECK

Why does the same indicated airspeed give a higher ground speed at a hot, high aerodrome?

A. The wind is stronger

B. Lower air density means a higher true airspeed for the same indicated airspeed

C. The engines produce more thrust

D. The runway is longer

Answer: B. At low density, a given indicated airspeed corresponds to a higher true airspeed, and therefore a higher ground speed.

Training material only. It does not replace the Aircraft Flight Manual, FCOM, QRH, MEL or the Domestic Airlines Operations Manual, which always take precedence.